Table 1: Popular programming platforms for RFID engineering and antenna design

Category	Type	Tool/Platform	Key role in RFID/ antenna design
EM (electromag-netic) simulation	3D EM solver	ANSYS HFSS	High-accuracy antenna and RFID tag simulation
EM simulation	EM solver	CST Studio Suite	Time/frequency domain antenna optimisation
EM simulation	EM solver	FEKO	Large-scale RFID system simulation
EM simulation	Multiphysics	COMSOL Multiphysics	RF + thermal coupled analysis
EM simulation	Time-domain solver	XFdtd	Radiation and SAR analysis
EM simulation	Open source EM	OpenEMS	Low-cost antenna prototyping
EM simulation	Planar EM	Sonnet Suites	Microstrip RFID antenna design
EM simulation	3D EM tool	EMPro	EM + circuit co-simulation
EM simulation	Lightweight EM	NEC (Numerical Electro-magnetics Code)	Wire antenna modelling
Circuit design	RF circuit tool	ADS (Advanced Design System)	RF front-end and co-simulation
Circuit design	Circuit simulator	LTspice	Analog RFID circuit design
Circuit design	IC design	Cadence Virtuoso	RFID chip/ASIC development
Circuit design	Circuit simulator	PSpice	SPICE-based RF simulation
Circuit design	Circuit tool	QUCS	RF circuit prototyping
System tools	System control	LabVIEW	RFID testing and automation
System tools	Network simulator	OMNeT++	RFID protocol simulation
System tools	Network simulator	NS-3	Wireless/RFID modelling
System tools	Security tool	RFIDIOT	RFID protocol testing
PCB design	PCB tool	Altium Designer	Professional RFID hardware design
PCB design	PCB tool	KiCad	Open source PCB prototyping
PCB design	PCB tool	Autodesk Eagle	Rapid prototyping
PCB design	PCB tool	OrCAD PCB Designer	Industry-grade PCB layout
PCB design	PCB + simulation	Proteus	Embedded RFID system design

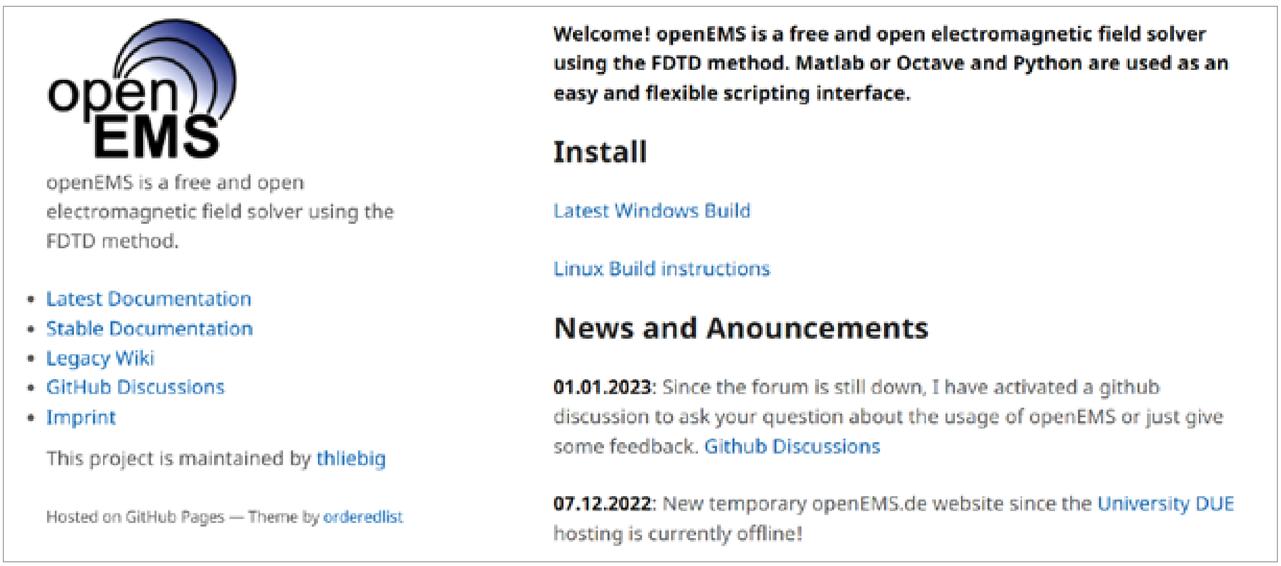

Figure 1: OpenEMS